

Chapter Five

Testing and Deployment

Web Application Testing

- Web application testing ensures the quality and performance of a web application before it is released to users.
- It involves checking the functionality, usability, security, compatibility, and performance of the application.

Key goals of web application testing:

- Validate that the application works as intended.
- Identify and fix bugs and issues.
- Ensure that the user experience is smooth across devices and browsers.
- Confirm that the application meets security and performance requirements.

Common web application components tested:

- Front-end (UI/UX)
- Back-end (API and database)
- Integration points (e.g., third-party services)
- Security features
- Data handling and form validations

Types of Testing

Functional Testing

- Verifies that the web application behaves according to the functional requirements.
- Examples: form submissions, login/logout, search, navigation.

Non-Functional Testing

- Focuses on how the system performs under certain conditions rather than what it does.
- Includes performance, usability, compatibility, and security testing.

Unit Testing

- Tests individual components or functions of the application in isolation.

Integration Testing

- Ensures different modules or services work well together.

Types of Testing

System Testing

- Tests the complete and integrated application.

Acceptance Testing

- Verifies the application meets the business requirements and is ready for deployment.

Regression Testing

- Ensures that new code changes have not adversely affected existing features.

Load and Stress Testing

- Assesses how the application performs under heavy traffic or usage.

Testing Tools

- Some commonly used tools for web application testing:

Testing Type	Tool Examples
Unit Testing	JUnit, NUnit, Mocha, Jasmine
Functional Testing	Selenium, Cypress, TestCafe
Performance Testing	JMeter, LoadRunner, Gatling
Security Testing	OWASP ZAP, Burp Suite
Cross-browser Testing	BrowserStack, Sauce Labs
Continuous Testing/CI	Jenkins, GitHub Actions, GitLab CI

Laboratory Contents

1. Functional Test Lab Activities

- **Objective:** Validate if the application performs its intended functions.
- **Examples:**
 - Testing form submissions and error handling.
 - Checking user authentication (login, logout).
 - Navigating through the web app to validate routing and page transitions.
 - Testing CRUD (Create, Read, Update, Delete) operations on data.

Laboratory Contents

Non-Functional Attributes Testing Lab Activities

- **Performance Testing:**
 - Simulating multiple users using tools like JMeter.
 - Measuring response time, throughput, and resource utilization.
- **Usability Testing:**
 - Collecting user feedback on UI/UX.
 - Measuring task completion rate and satisfaction.
- **Compatibility Testing:**
 - Testing the application on different browsers (Chrome, Firefox, Safari).
 - Running the app on various devices (desktop, tablet, mobile).
- **Security Testing:**
 - Scanning for vulnerabilities using OWASP ZAP or similar tools.
 - Testing input validation to prevent SQL injection, XSS, CSRF.